C3.1 Integration of body systems

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c3-1.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Image credits are on the last slide of the slides for this subtopic.

WORKING TOGETHER

C3.1.1 What it means for systems to be integrated

p. 491

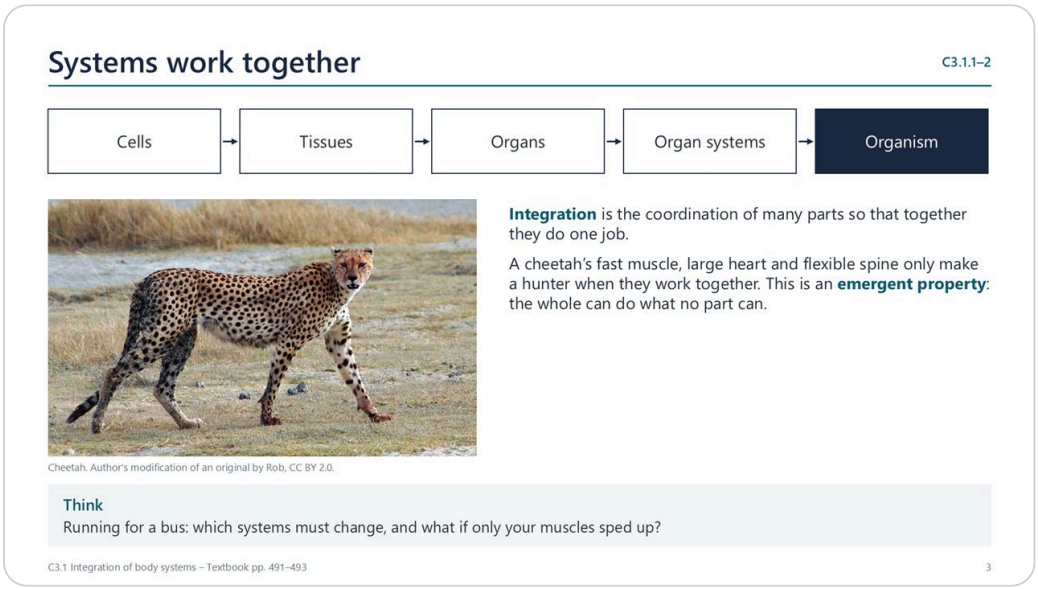
Integration is the coordination of the parts of a system, so that together they perform an overall function. It is necessary at every level of life, because the parts would otherwise work against one another.

C3.1.2 The hierarchy from cells to systems

pp. 491–493

Cells form tissues, tissues form organs, organs form organ systems, and organ systems form the organism. New properties appear at each level. These **emergent properties** arise from interaction, and the parts alone do not have them.

A cheetah is an effective predator only because its muscles, spine, heart and lungs work together.



C3.1.3 How hormones, nerves and transport link organs

pp. 493–495

	NERVOUS	HORMONAL
SIGNAL	Electrical impulses and neurotransmitters	Chemical messengers
TRAVELS BY	Neurons and nerves	Bloodstream
DESTINATION	Specific muscles or glands	Target cells around the body
SPEED AND DURATION	Very rapid, short-lived	Can be slow, long-lasting

The **blood system** also integrates organs by moving materials. Glucose goes from the gut and liver to respiring cells, oxygen from the lungs to all cells, and urea from the liver to the kidneys.

THE NERVOUS SYSTEM

C3.1.4 The brain as the main centre for integrating information

p. 496

The brain receives impulses from sensory receptors, combines and processes information from several inputs at once, and sends impulses to effectors. It also stores information, in **learning and memory**.

C3.1.5 What the spinal cord coordinates without conscious thought

p. 498

The spinal cord carries impulses between the brain and the body. It also coordinates **unconscious** processes, which are automatic, such as reflexes. **Conscious** processes, such as voluntary movement and speech, are deliberate. Its grey matter contains cell bodies and synapses, and its white matter contains myelinated fibres.

C3.1.6 How sensory neurons carry input

pp. 498–499

Sensory neurons carry messages from receptor cells to the central nervous system: to the spinal cord and on to the cerebral hemispheres.

C3.1.7 How motor neurons carry output

p. 499

Motor neurons carry impulses from the cerebral hemispheres, through the spinal cord, to muscles, which are stimulated to contract.

C3.1.8 Nerves: bundles of sensory and motor fibres

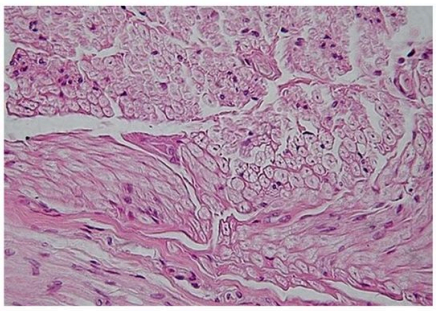
p. 500

A nerve is a bundle of nerve fibres from both sensory and motor neurons, inside a protective sheath. A transverse section shows both myelinated and unmyelinated fibres.

Watch out: a neuron is one cell, and a nerve contains fibres from many neurons.

Nerves are bundles of nerve fibres

C3.1.8



A peripheral nerve in section. CC BY-SA 3.0.

A **nerve** holds fibres from both **sensory and motor neurons**, grouped in bundles inside a protective sheath.

Most nerves have both **myelinated** and **unmyelinated** fibres.

Common mistake

A neuron is one cell. A nerve is a bundle of many fibres from many neurons.

C3.1 Integration of body systems – Textbook p. 500 11

C3.1.9 The pain reflex arc

pp. 501–502

The pain reflex arc

C3.1.9

Stimulus: heat → **Pain receptor**: free nerve ending → **Sensory neuron** → Interneuron in grey matter → **Motor neuron** → **Effector**: skeletal muscle

A **reflex** is a rapid, **involuntary** response to a stimulus. The hand moves before you feel the pain, because the signal does not need to reach the brain first.

Other fibres do carry the message to the brain, so you become aware of it a moment later.

Exam note

Learn this example: free nerve ending in the hand, one interneuron in the grey matter of the spinal cord, skeletal muscle as the effector.

C3.1 Integration of body systems – Textbook pp. 501–502 14

A reflex is a rapid, involuntary response. A free nerve ending in the hand, acting as a pain receptor, starts an impulse in a **sensory neuron**. This passes to one **interneuron** in the grey matter of the spinal cord, then to a **motor neuron**, and then to skeletal muscle, which is the effector.

C3.1.10 How the cerebellum coordinates movement and balance

p. 502

The cerebellum, in the hindbrain, coordinates skeletal muscle contraction and balance. It uses input from receptors to adjust commands to motor neurons. It refines movements, but does not start them.

HORMONES

C3.1.11 How melatonin adjusts sleep

pp. 502–503

The **pineal gland** secretes melatonin, more in darkness and less in light. High levels promote sleep, and falling levels prepare the body to wake. This helps set the 24-hour **circadian rhythm**. Jet lag happens because the rhythm takes days to adjust.

C3.1.12 How epinephrine readies the body for activity

pp. 503–504

The **adrenal glands** secrete epinephrine (adrenaline) when the body is under stress. Heart rate and breathing rate rise, and liver glycogen is broken down, so more oxygen and glucose reach muscles. Aerobic respiration then supplies more ATP for intense contraction.

C3.1.13 How the hypothalamus and pituitary direct the glands

pp. 504–505

The **hypothalamus** links the nervous and endocrine systems. It monitors the blood, and it controls the **pituitary gland** using hormones and nerve impulses. Pituitary hormones then control growth, development, reproduction and homeostasis.

FEEDBACK

C3.1.14 Feedback control of heart rate

pp. 506–507

Baroreceptors in the aorta, carotid arteries and right atrium monitor blood pressure. **Chemoreceptors** in the carotid arteries, aorta and brain monitor blood pH and levels of oxygen and carbon dioxide. The **medulla** coordinates the response.

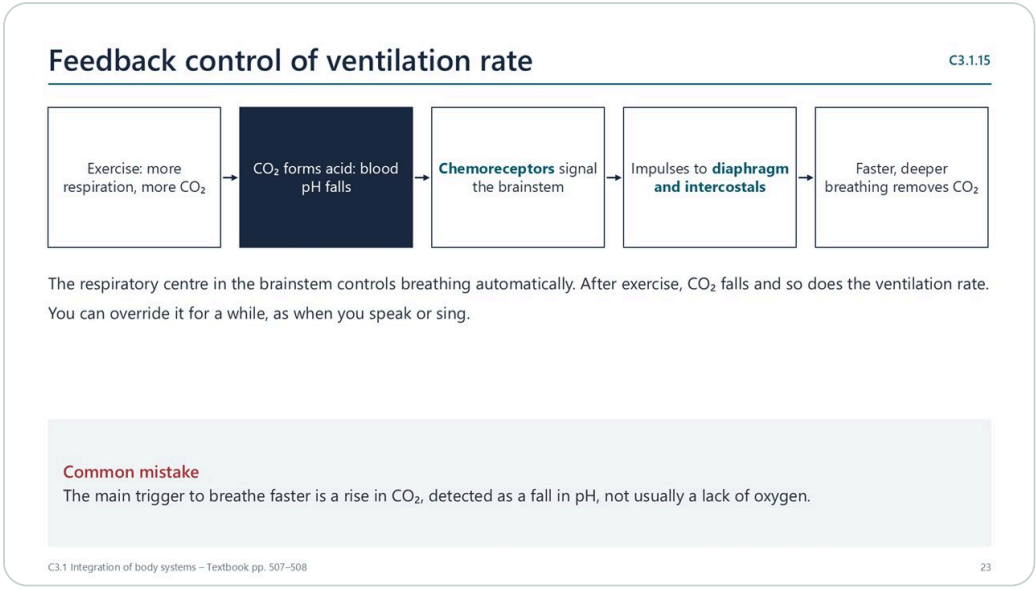
DETECTED	NERVE TO THE SINOATRIAL NODE	RESULT
LOW PRESSURE, LOW PH OR LOW OXYGEN	Sympathetic	Heart rate rises
HIGH PRESSURE, OR LOW CARBON DIOXIDE	Vagus (parasympathetic)	Heart rate falls

C3.1.15 Feedback control of ventilation rate

pp. 507–508

During exercise, respiration produces more carbon dioxide, which forms an acid in the blood, so pH falls. **Chemoreceptors** detect this and signal to the brainstem. Impulses go to the diaphragm and intercostal muscles, so breathing is faster and deeper, and carbon dioxide is removed.

Watch out: the main stimulus is a rise in carbon dioxide, detected as a fall in pH, and not usually a lack of oxygen.



C3.1.16 How the central and enteric nervous systems control peristalsis

pp. 508–509

Peristalsis is waves of muscle contraction along the gut. Swallowing and egestion are voluntary, under the central nervous system. Between these two points, movement is involuntary and is coordinated by the **enteric nervous system** in the gut wall.

© 2026 Akash Parthi Bendran. Please ask before using or sharing: see whatthecell.org/about.html#permission